LIMO PROCESSOR

M2 INTERFACE WIREFRAMES

A visual plan for the processor simulator interface

01 Main Simulator Workspace

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SOURCE CODE

[Open file] [Save file]

```
┌──────────────────────────────────────────────┐
│ 1   [ Enter assembly instruction ]           │
│ 2   [ Enter assembly instruction ]           │
│ 3   [ Enter assembly instruction ]           │
│ 4   ...                                      │
└──────────────────────────────────────────────┘
```

EXECUTION CONTROLS

[Assemble]

[Step]

[Run]

[Pause]

[Reset]

STATUS

THE CONTROL UNIT

[Ready / error / success]

MACHINE CODE / ASSEMBLER RESULT	
Address	Machine-code output
0x0000	[encoded instruction]
0x0004	[encoded instruction]
0x0008	[encoded instruction]

02 Pipeline and Datapath View

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IF	ID	EX	MEM	WB
Instruction Fetch Current instruction: [value / empty]	Instruction Decode Current instruction: [value / empty]	Execute Current instruction: [value / empty]	Memory Access Current instruction: [value / empty]	Write Back Current instruction: [value / empty]

PIPELINE STATUS

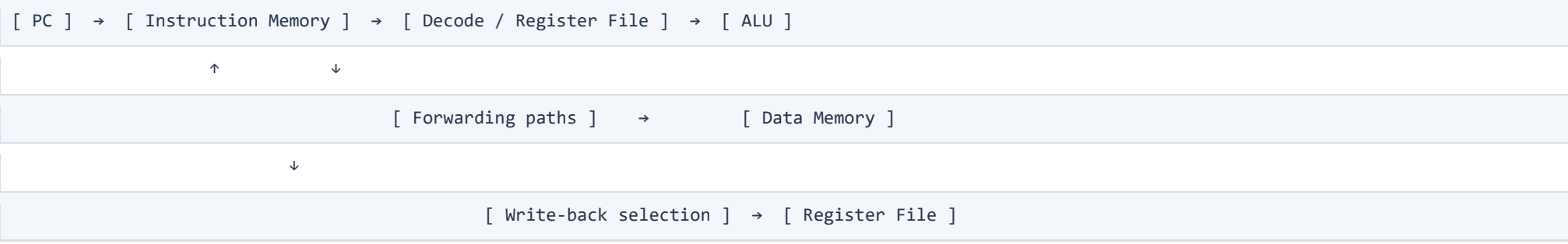
Current cycle: [number] Program counter (PC): [value]

Hazard detected: [Yes / No] Stall: [Yes / No] Flush: [Yes / No]

Forwarding source: [None / EX-MEM / MEM-WB]

Branch decision / next PC: [status or value]

DATAPATH CONCEPT



03 Register and Memory Inspector

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ARCHITECTURAL REGISTERS		MEMORY INSPECTOR	
r0	[register value]	0x0000	[stored value]
r1	[register value]	0x0004	[stored value]
r2	[register value]	0x0008	[stored value]
r3	[register value]	0x000C	[stored value]
r4	[register value]	0x0010	[stored value]
r5	[register value]	0x0014	[stored value]
r6	[register value]	0x0018	[stored value]
r7	[register value]	0x001C	[stored value]

Memory view: [Instruction memory / Data memory] **Cycle:** [number]

04 Pipeline Register and Control Inspector

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IF/ID REGISTER

- instruction: [value]
- pc: [value]
- pcPlus4: [value]

ID/EX REGISTER

- pc / pcPlus4: [value]
- source registers: [rs1] [rs2]
- destination: [rd]
- operand values / immediate: [values]
- control fields: [values]

EX/MEM REGISTER

- ALU result / store data: [values]
- destination and memory/write-back controls: [values]

MEM/WB REGISTER

- memory/ALU result selected for write-back: [value]
- destination / RegWrite: [values]

05 Execution Trace and Messages

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A readable history of cycles, stage activity, and important events.

EXECUTION SUMMARY

[Step] [Run] [Pause] [Reset]

Current PC: [value] Cycle count: [number]

CYCLE-BY-CYCLE TRACE

Cycle	IF	ID	EX	MEM	WB	Event
001	[instr]	[instr]	[empty]	[empty]	[empty]	[status]
002	[instr]	[instr]	[instr]	[empty]	[empty]	[status]
003	[instr]	[instr]	[instr]	[instr]	[empty]	[status]
004	[instr]	[instr]	[instr]	[instr]	[instr]	[status]

OUTPUT / ASSEMBLER MESSAGES

[Assembly success, warning, execution message, or error appears here.]

[Program output appears here if the simulator supports program output.]